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Ethical and Legal Analysis of AI in Biotech: A Comprehensive Business Assessment for PARACELUS in the European Union

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Prepared by:

Amaan Mohammed Khalander / Q1132588

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AMAAN MOHAMMED KHALANDER

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Executive Summary

Paracelsus proposed services create major ethical and legal exposure because they combine genetic data, health inference, personality profiling, aptitude prediction, and psychological interpretation. In the EU, genetic and health data are special-category data under GDPR Article 9 and Recital 34, and employee consent is especially fragile because the imbalance of power can make voluntary participation difficult to treat as freely given.

The strongest product case is GeneGuard, provided it is positioned conservatively as a probabilistic risk-support tool with clear explanations and clinician-aligned communication. PolyRisk may also be viable, but only if Paracelsus can demonstrate robust ancestry-stratified validation and avoid overstating accuracy across populations, because polygenic tools have been shown to perform less well outside European ancestry samples.

PersonaMap is ethically questionable and legally risky if used to infer personality traits from genetic data, while AptiCore and PsycheDecode are the most problematic, especially in B2B settings where they could influence employment-related decisions. Under the EU AI Act, employment-related uses such as recruitment, task allocation, monitoring, and performance evaluation may trigger high-risk obligations, while workplace emotion-related inference raises additional compliance concerns.

The recommended strategy is to retain GeneGuard, constrain PolyRisk, heavily redesign or pause PersonaMap, and withdraw AptiCore and PsycheDecode from employer-facing use. A strong governance model covering DPIAs, bias testing, model documentation, audits, transparency reporting, and an internal ethics board is essential if Paracelsus wants to operate credibly in the EU.

Introduction

Paracelsus is a fictitious biotech sequencing company planning to launch five AI-driven genetic services that predict disease risk, personality tendencies, aptitude, and psychological traits. This report examines the ethical and legal implications of these services in the European Union, focusing on privacy, fairness, autonomy, transparency, and accountability. It also evaluates the GDPR and the EU AI Act, before recommending practical mitigation measures and governance steps to support compliant and responsible deployment.

Paracelsus operates in a domain where technical capability alone does not justify deployment, because genetic AI can shape decisions about health, identity, and opportunity in ways that are difficult to reverse. Floridi and Cowls (2019) argue that AI systems must be evaluated against five core principles, namely beneficence, non-maleficence, autonomy, justice, and explicability, and that all five must be satisfied simultaneously rather than traded off against one another. This makes the Paracelsus portfolio particularly challenging to assess, given that different products satisfy these principles to very different degrees. In the European Union, this matters especially because genetic and health data are treated as special-category data under GDPR Article 9 and Recital 34 (European Union, 2016), and AI systems used for profiling or employment-related decisions may trigger additional duties under the EU AI Act (European Union, 2024). This report therefore assesses not only whether the proposed products are lawful, but also whether they are ethically defensible in light of autonomy, fairness, transparency, accountability, and the likely real-world effects on consumers and employees.

Overview of Ethical and Legal Issues

Paracelsus will process exceptionally sensitive personal data, including genetic, health, and inferred psychological information. This creates privacy risks because genetic data are inherently identifying, long-lived, and potentially informative about relatives as well as the individual, so a breach can have broader and more permanent consequences than an ordinary data leak (Aicardi et al., 2016).

The stakeholder map is broader than customers and employers. Individual consumers may benefit from earlier risk awareness, but they may also experience anxiety, confusion, or false certainty if probabilistic outputs are presented too confidently. Employees are exposed to a much sharper imbalance of power, because even “voluntary” participation can be shaped by managerial pressure, fear of exclusion, or concern that non-participation may be viewed negatively. Employers may receive only aggregated results, but aggregated insights can still lead to proxy discrimination, subtle workforce sorting, or function creep if the data are later reused for purposes that employees did not expect.

The five products are ethically unequal. GeneGuard is the most defensible because it can be framed as health-supportive risk information, whereas PolyRisk raises fairness concerns if its accuracy differs across ancestry groups. PersonaMap, AptiCore, and PsycheDecode are more problematic because they extend from disease-related inference into personality, aptitude, and psychological profiling, which increases the danger of stigma, over-interpretation, and misuse in contexts such as employment or education. In other words, the ethical risk increases as the product moves further away from health support and closer to social classification.

Bias and fairness are central concerns because AI systems trained on skewed genetic datasets can reproduce unequal performance across ancestry groups. Research on polygenic risk scores shows that current models often perform best in individuals of European ancestry and can worsen health disparities if deployed without careful recalibration and population-specific validation (Martin et al., 2019). That means PolyRisk is not just a technical product issue; it is a distributive justice issue. The WHO (2021) reinforces this concern, noting that AI health tools must be evaluated not only for average performance but for whether they widen or narrow existing disparities across populations, making equitable validation a precondition for responsible deployment.

Accountability and transparency are also essential because customers and regulators need to understand what each system does, what it cannot do, and who is responsible when outputs are inaccurate or harmful. Products like PersonaMap, AptiCore, and PsycheDecode are particularly sensitive because they move from health prediction into identity, capability, and behaviour inference, which increases the risk of overclaiming scientific certainty and creating unjustified trust in the outputs.

In-Depth Ethical Analysis

From a deontological perspective, Paracelsus must treat users as ends in themselves, not as raw material for commercial profiling. That principle is strained by services that infer personality, aptitude, or psychological traits from genetic data, because those outputs can reduce a person to a set of probabilistic labels rather than respect them as autonomous agents with contextual, social, and moral complexity.

A useful ethical framework here is principlism (Beauchamp and Childress, 2019). Respect for autonomy requires that users understand what the system can and cannot tell them, while beneficence requires Paracelsus to produce genuine value rather than commercial novelty. Non-maleficence is especially important because inaccurate or overconfident genetic interpretation can cause distress, unnecessary follow-up testing, or harmful lifestyle decisions based on misunderstood risk. Justice is central to PolyRisk because a model that performs better for some ancestry groups than others does not merely reflect technical imperfection; it risks reproducing existing health inequalities (Beauchamp and Childress, 2019).

Deontological ethics also supports a strict limit on the most speculative products. If Paracelsus sells personality or aptitude inference as though it were settled science, it risks treating individuals as instruments for prediction and sorting rather than as persons with context, agency, and dignity. That problem is especially serious for employer-facing use, because even aggregate outputs can shape how people are perceived inside a workplace. A company could justify GeneGuard under a beneficence-based argument, but that same argument is far weaker for PsycheDecode or AptiCore, where the practical benefit is uncertain and the possibility of harm is much clearer.

Biomedical ethics provides a more practical framework. GeneGuard may support beneficence by helping users understand disease risk, but non-maleficence requires careful communication to avoid panic, false reassurance, or unnecessary downstream spending. A realistic harm scenario would be a user interpreting a high-risk result as a diagnosis and making major life or treatment decisions on the basis of a probabilistic estimate. Beauchamp and Childress (2019) stress that disclosure of probabilistic medical information must be accompanied by adequate explanation of uncertainty and appropriate referral pathways, precisely to prevent this kind of interpretive harm.

Fairness concerns are especially serious for PolyRisk. The ethical issue is not simply whether the model is accurate, but whether the benefits and burdens are distributed fairly across populations, since ancestry-skewed training data can systematically privilege some users over others. That is why deploying PolyRisk without subgroup validation would be ethically indefensible even if the model performs well on average (Martin et al., 2019). Floridi and Cowls (2019) similarly argue that justice in AI requires not merely avoiding overt discrimination but actively ensuring that benefits are distributed fairly, which means ancestry-stratified validation is not optional but a minimum ethical requirement for any polygenic service.

Utilitarian reasoning can support some Paracelsus services if they produce meaningful health gains at scale, but the utility case collapses where outputs are weak, confusing, or socially harmful (Floridi and Cowls, 2019). PsycheDecode and AptiCore are ethically hardest to defend because they can generate self-fulfilling labels about capability or mentality, especially if employers begin to treat aggregated outputs as quasi-scientific workplace intelligence.

A realistic harm scenario is not limited to technical failure. A user may interpret a high-risk score as a diagnosis, or an employer may treat aggregate aptitude profiles as a legitimate basis for workforce planning, even where the product was not designed for that purpose. These are classic examples of overreach and misplaced trust. From a utilitarian perspective, a product only creates net benefit if its gains are real, its harms are limited, and its use is predictable; for Paracelsus, that standard is most plausible for GeneGuard, conditional for PolyRisk, and weak for the remaining services

In-Depth Legal Analysis (EU Jurisdiction)

Under GDPR, Paracelsus must identify both a lawful basis under Article 6 and a specific Article 9 condition for processing genetic and health data, which are special categories of personal data (European Union, 2016). For consumer use, explicit consent may be available, but it must be freely given, informed, specific, and easy to withdraw in practice. That means the company cannot hide broad secondary uses inside a general privacy policy, and it cannot bundle unrelated purposes into a single vague permission request.

Because Paracelsus processes genetic and health data at scale, a Data Protection Impact Assessment should be treated as an essential pre-launch control rather than a formality. A DPIA would help identify risks to individuals, test whether consent is genuinely valid, document safeguards, and show that the company has considered the consequences of profiling before deployment. This is particularly important where sensitive data, predictive modelling, and possible cross-context reuse are involved.

The employee scenario is legally more fragile. The EDPB makes clear that consent in employment is usually not freely given because of the imbalance of power between employer and employee (European Data Protection Board, 2020), so Paracelsus should not rely on employee consent as its main legal basis for workplace deployment. Even if only aggregated findings are shared with employers, the initial processing of highly sensitive genetic data still occurs, and that may create legal exposure if participation is not genuinely voluntary or if employees fear consequences for refusing. In practice, this makes B2B deployment much more difficult to defend than a direct-to-consumer model.

The EU AI Act further raises the compliance burden. Annex III identifies employment-related AI uses, including recruitment, task allocation, monitoring, and performance evaluation, as high-risk contexts (European Union, 2024), which means AptiCore is especially exposed if it is used for career guidance or workplace decision support. PsycheDecode is even more sensitive because psychological or emotional inference in workplace settings can drift into prohibited or heavily restricted territory, particularly if the output is used to judge suitability, stability, or performance. That means Paracelsus cannot treat legal compliance as a box-ticking exercise; it needs product restrictions that reflect the actual risk profile of each service.

The current legal framework is strong on data protection and risk classification, but it does not fully resolve scientific validity, manipulative product framing, or the ethics of inferential profiling. That means legal compliance alone is not enough; Paracelsus must apply a higher internal standard than the minimum required by law. Aicardi et al. (2016) note that digital health data governance has consistently lagged behind technological capability, leaving gaps that companies must fill through internal policy rather than waiting for regulatory reform. This is especially relevant to Paracelsus, whose services span a spectrum from clinically grounded disease-risk tools to speculative psychological profiling, requiring differentiated governance responses rather than a uniform compliance approach.

There is also a gap between formal compliance and substantive legitimacy. GDPR and the AI Act regulate data handling, risk classification, and governance, but they do not fully solve scientific uncertainty, misleading marketing, or the social consequences of turning probabilistic inference into business tools. That is important because a model can be lawful and still be ethically weak if it encourages overconfidence or produces unequal outcomes across populations. For this reason, the safest legal strategy is also the most ethically cautious strategy: limit deployment, narrow use cases, and avoid unsupported claims.

Mitigation Strategies

Paracelsus should adopt a tiered product strategy rather than a one-size-fits-all launch model. GeneGuard can be retained, but only with conservative health framing, plain-language uncertainty statements, and referral guidance that prevents users from treating predictions as diagnoses. PolyRisk should be limited to validated populations and supported by ancestry-stratified testing, periodic recalibration, and a published explanation of when the model is unreliable. PersonaMap should either be redesigned into a low-stakes exploratory product or paused until Paracelsus can justify its scientific basis more convincingly.

AptiCore should be removed from employer-facing use unless it is fundamentally redesigned to avoid recruitment, promotion, task allocation, or capability ranking. Even then, the company should assume a high-risk compliance burden because employment contexts amplify bias and consent problems. PsycheDecode should be excluded from B2B deployment altogether, because psychological or emotional inference is difficult to justify ethically and creates substantial legal exposure when used in workplace settings.

At the technical level, Paracelsus should use privacy by design measures such as encryption, pseudonymisation, role-based access control, strict retention limits, and separation of raw genomic data from derived reports. It should also maintain model cards and data sheets that document training data provenance, demographic coverage, intended purpose, limitations, and known failure modes. External bias audits, incident response procedures, and a cross-functional ethics board with authority to pause launches would help move the company from minimal compliance to credible governance (World Health Organization, 2021).

Recommendations

In operational terms, Paracelsus should implement a launch gate for every product, requiring legal review, ethics review, and technical bias testing before any public release. It should also publish a concise transparency notice for consumers that explains what each product does, what it does not do, and how users can challenge or withdraw their data use. For employer-facing products, the company should prohibit access to individual results, prevent secondary use, and ensure that aggregated reporting cannot be used for hidden personnel decisions.

1. Retain GeneGuard with tight clinical framing. Market it as a probabilistic risk-support tool, not a diagnostic product, and use plain-language explanations for uncertainty and limitations.
2. Restrict PolyRisk to validated populations. Require ancestry-stratified performance testing, periodic recalibration, and disclosure of where the model is unreliable.
3. Redesign PersonaMap or suspend it. Do not present personality inference from genetic data as scientifically settled; if retained, it should be exploratory, user-controlled, and non-diagnostic.
4. Withdraw AptiCore from employer use. Employment-related aptitude inference is too close to high-risk profiling and creates serious consent and fairness problems.
5. Exclude PsycheDecode from B2B deployment. Psychological or emotional inference in the workplace is ethically fragile and legally exposed.
6. Adopt consent and transparency redesign for consumers. Separate each product and purpose, make withdrawal easy, and explain results in a way ordinary users can understand.
7. Create independent oversight. Use periodic ethics review, third-party audits, and public transparency reporting to demonstrate accountability and build trust.

Paracelsus should also adopt a clear stakeholder communication plan before launch. Consumers need simple explanations of what the services can and cannot infer, while employees need explicit assurance that non-participation will not affect recruitment, promotion, or evaluation. This matters because trust in genetic AI depends not only on technical safeguards but also on whether people believe the company is being honest about uncertainty, limits, and downstream use. A short public transparency notice and an internal employee notice would strengthen both legal compliance and commercial credibility.

Concluding Remarks

Paracelsus can pursue responsible innovation in the EU only if it accepts that not every technically possible product is ethically or legally suitable for market. GeneGuard and a carefully governed PolyRisk are the most defensible offerings, while PersonaMap requires major redesign and AptiCore and PsycheDecode should be removed from employer-facing deployment.

The central lesson is that genetic AI is not just a data-processing problem; it is a fundamental-rights issue involving dignity, fairness, autonomy, and accountability. If Paracelsus builds its strategy around limitation, validation, transparency, and governance, it can create trust and reduce regulatory exposure while preserving the most credible parts of its portfolio.

Overall, the assessment shows that Paracelsus should not treat all five products as equally viable. The safest path is to preserve the health-supportive products, tightly constrain any predictive risk scoring, and avoid moving into speculative personality or workplace profiling unless the legal basis, scientific validity, and ethical justification are all much stronger. Responsible innovation here means choosing the products that can be defended, rather than expanding the portfolio simply because the technology makes it possible.

Acknowledgement

Perplexity AI was used in the preparation of this report to support literature searching, identify potentially relevant academic sources, and assist with grammar correction and word-limit reduction during editing. All sources included in the reference list were checked by the author and cited as the original academic works rather than citing Perplexity AI as a source. The analysis, interpretation, discussion, and final written content submitted in this report are the author's own work and responsibility.

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